Unit no. ° 15

CONSOLIDATION AND TRANSITION TO THE FIRST

25 May·11 June 2027

I Calculation balance, functions and geometry

Example 15.1: Annual Review · Part A

1. Make 168 /252 irreducible; calculate $2^{-3} + 1/4$.

2. Factoring $x^2 - 6x + 9$, then solve $(x - 3)^2 = 4$.

3. Solve $\dfrac{x-2}{x+1} \geq 0$.

4. For $f(x) = 4 - (x-1)^2$ on [−2; 3], give the maximum and the apses where $f(x) = 0$.

5. With $A(0;1)$, $B(4;3)$, C(6; 4), demonstrate alignment and give a straight line equation.

Solution:

1. $168/252 = 2/3$, and $2^{-3} + 1/4 = 1/8 + 2/8 = 3/8$. 2. $(x-3)^2$; $x - 3 = \pm 2$, so $S = \{1; 5\}$. 3. Forbidden -1, zero 2; signs + , −, +, so $S =]-\infty; -1[\cup [2; +\infty[$. 4. The square is positive or null, so $f(x) \leq 4$, equality in x = 1. The zeros are $x = -1$, 3, both in the field. 5. $\overrightarrow{AB} = (4; 2)$, $\overrightarrow{AC} = (6; 3) = \frac{3}{2}\overrightarrow{AB}$. An equation is $x - 2y + 2 = 0$.

Benchmarks for the teacher

Indicative scale: 10 points, 2 points per question.
For question 5, request the conclusion after the calculation.